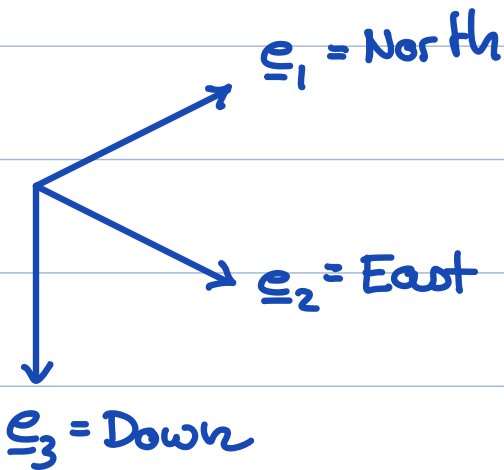


Fault normals from dip and strike

Geographic coordinate system (NED)



need to add discussion
about the direction of
the dip?

we only need two numbers
because vector is unit length

Geological description of a fault

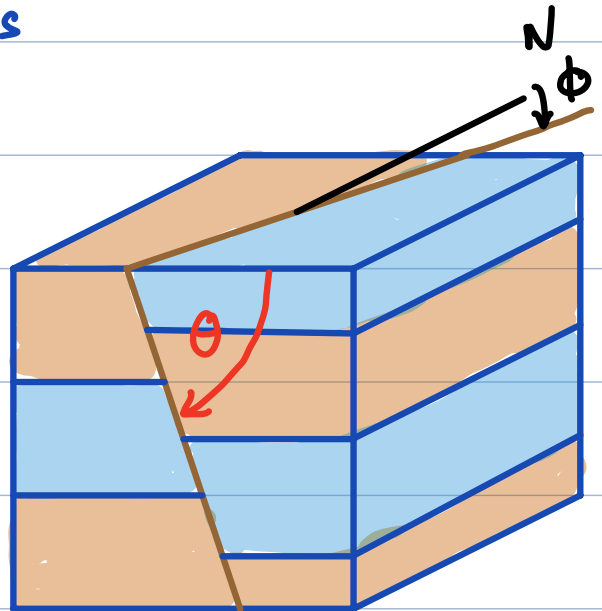
Fault is a discontinuity cross-section view
in the Earth across which rocks
have been displaced.

Faults are generated by
earth quakes.

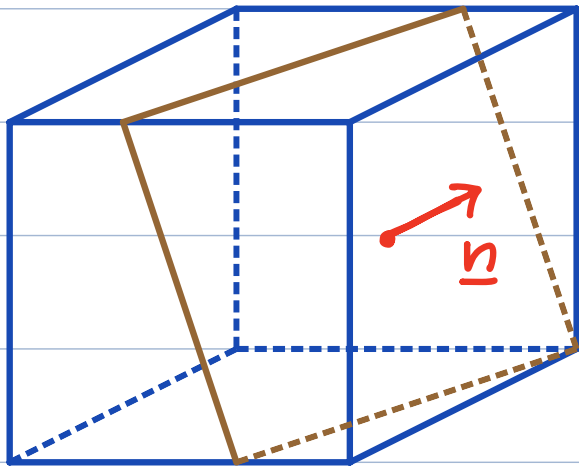
Orientation is given by
dip and strike angles.

ϕ = strike = angle from north

θ = dip = angle from horizontal
(perpendicular from strike)



Q: Given ϕ & θ what is normal to fault plane?



Start with $-\underline{e}_3$

Two rotations:

1) Rotation around $\underline{e}_1$

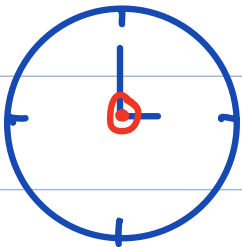
by dip: $\underline{Q}_D = \underline{Q}_1(\underline{e}_1, \theta)$

2) Rotation around $\underline{e}_3$

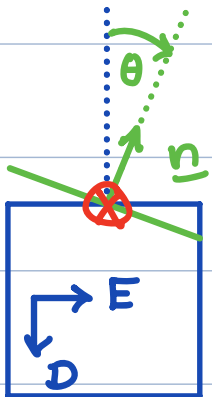
by strike: $\underline{Q}_S = \underline{Q}(\underline{e}_3, \phi)$

$$\underline{n} = \underline{Q}_S \underline{Q}_D (-\underline{e}_3)$$

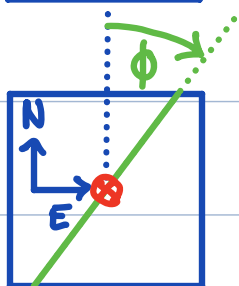
Note, both are counterclockwise rotations!



standard definition assumes
rotation axis points out of page.



dip rotation: $\underline{e}_1$ points into page
 $\Rightarrow$ counter clockwise



strike rotation: $\underline{e}_3$ points into page
 $\Rightarrow$ counter clockwise

General rotation matrix

$$\underline{\underline{Q}}(\underline{r}, \theta) = \underline{r} \otimes \underline{r} + \cos \theta (\underline{\underline{I}} - \underline{r} \otimes \underline{r}) + \sin \theta \underline{\underline{R}}$$

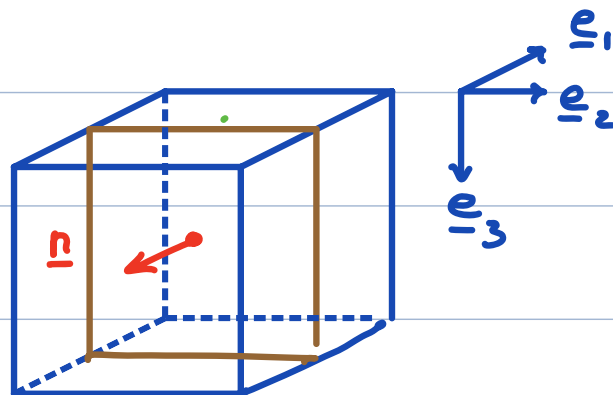
Simple example:

$$\text{dip: } \theta = 90^\circ \left(\frac{\pi}{2}\right)$$

$$\text{strike: } \phi = 90^\circ \left(\frac{\pi}{2}\right)$$

by inspection: $\underline{n} = -\underline{e}_1$

$$\sin\left(\frac{\pi}{2}\right) = 1 \quad \cos\left(\frac{\pi}{2}\right) = 0$$



$$\underline{\underline{R}} = \begin{bmatrix} 0 & -r_2 & r_2 \\ r_3 & 0 & -r_1 \\ -r_2 & r_1 & 0 \end{bmatrix}$$

Dip rotation:

$$\underline{\underline{Q}}_D = \underline{e}_1 \otimes \underline{e}_1 + \underline{\underline{R}}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

Strike rotation:

$$\underline{\underline{Q}}_S = \underline{e}_3 \otimes \underline{e}_3 + \underline{\underline{R}}_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\underline{\underline{Q}}_D$ and $\underline{\underline{Q}}_S$ are simple because 90° rotations around coordinate axes.

Compute fault normal:

$$\underline{n} = \underline{Q}_s \underline{Q}_D (-\underline{e}_3)$$

$$\underline{n} = \underline{Q}_s \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} = \underline{Q}_s \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\underline{n} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} = -\underline{e}_1 \quad \checkmark$$

Note: Tensor multiplication does not commute!

$$\underline{Q}_s \underline{Q}_D \neq \underline{Q}_D \underline{Q}_s$$

The order of rotation matters